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A survey of deep learning-driven architecture for predictive maintenance

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ABSTRACT

Over the past decades, deep learning techniques have attracted increased attention from various research and industrial domains aligned with the development of Industry Internet-of-Things(IIoT). Specifically, with the advantage of data-driven methods, industrial organizations are seeking novel proactive strategies supported by analytic models to guarantee the quality of their production by observing degradation or predicting failure ahead of the occurrence of the component or asset. Predictive strategies are expected to promise the influence of unnecessary maintenance interruptions and mitigate the consequence of that, hence, extending the remaining useful life of products. This paper conducts a survey of the utilization of deep learning technologies on engineering applications where they provide satisfactory solutions with respect to specific data types or input signals. 106 primary papers are reviewed on deep learning–driven approaches which mainly explore five of the most popular architectures in the application of predictive maintenance. The main content of this paper summarizes the common advantages of each architecture and, accordingly, points out their limitations, as well as describes the application scopes of fully connected deep neural networks, convolutional neural networks, stacked autoencoders, deep belief networks, and deep recurrent neural networks. Based on the technique discussion for each of them, we intend to provide a comprehensive understanding and guidance of the appropriate usage of deep learning architectures to devise an effective predictive maintenance strategy for the scientific and industrial developers whose expertise lies in the prior domain knowledge of multi-source isomerization data. Moreover, the main content demonstrated the summarization of the decisive factor by which the incremental stages of the approaches were determined, fundamentally including the dataset specification, feature extraction, and the integration of deep learning approaches.

1. Introduction

Over the past decades, the concept of a fourth industrial revolution, also known as Industry 4.0, has attracted great prominence and is now a top priority of organizations and researchers (Ghobakhloo, 2018). Concurrently, the concept is also sought for its potential to leverage the adoption of a circular economy (Lopes de Sousa Jabbour et al., 2018)—a promising concept to promote sustainable development (Ghisellini et al., 2016). This can be achieved by increasing the material resource efficiency and productivity through the support, or unburden, of human decision-making in strategies such as maintenance. However, maintenance remains a challenging task for manufacturing companies and is one of the most essential strategies for reducing unplanned downtime and loss of productivity. A study by the Wall Street Journal and Emerson (2018) estimated the cost of unplanned downtime for industrial manufacturers to be 50 billion USD each year. Equipment or system failure was seen as the root cause of 42% of this downtime.

Furthermore, equipment failure and unplanned outages result in excessive material use and cost and expose factory workers to a greater safety risk. To address this challenge, companies have adopted preventive maintenance actions by scheduling equipment inspections and replacements on a regular and timely basis. However, with preventive maintenance, parts are usually replaced too early and consequently carry an unnecessary financial burden and material waste for the company. According to IBM and a study by ARC Advisory Group (Bunzel, 2016), the average company can reduce its cost of preventive maintenance by up to 50% by optimizing the maintenance frequency following the actual equipment's condition. With the development of IoT and artificial intelligence (AI) technologies, it is possible to collect and analyze accurate sensory data to identify previously hidden patterns of

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behavior for more precise identification and prediction of potential failures and degradations. The so-called predictive maintenance strategy supports a more circular and sustainable economy by determining when to replace certain parts, thereby reducing downtime and material use, increasing utilization, extending the remaining useful life (RUL), and enabling products to loop through additional use cycles because of improved insight into condition and usage history (Kristoffersen et al., 2019).

Returning to the fundamental issue, the advanced structure of deep neural networks, such as convolution neural networks, has presented the robust capability of extracting both time-domain and frequency-domain information instead of applying traditional signal processing techniques with manual configuration (Tao et al., 2023). This aspect was commonly considered as the indispensable factor that a huge amount of engineers and researchers express their willingness to apply CNN together with the information fusion mechanism to enhance the performance of the deep learning approach(Xu et al., 2023a). As they mentioned, the integration of signal processing techniques, such as wavelet transform, empirical mode decomposition, and Fourier transform, and the deep neural network achieved great success in recognizing the degradation pattern from raw signals. Alternatively, several state-of-the-art frameworks such as transfer learning(Zhang and Li, 2021; Zhang et al., 2021), Q-learning (Wang et al., 2023), and physics-informed architectures (Raissi et al., 2019; Sadoughi and Hu, 2018; Xu et al., 2023b; Zhu et al., 2023) were proposed to improve the reliability and sustainability of data-driven predictive maintenance applications. The motivation of applying Q-learning and transfer learning addresses the drawback of the existing challenge of insufficient offline training dataset, which consequently leads to the ineffectiveness of the trained model. These advances allow the encouragement from the external working conditions to estimate the unknown failure in terms of stochastic operation conditions, meanwhile fine-tuning the deep learning model which contains millions of parameters. Some latest scholars also reported that the intrinsic structures of the deep learning architecture are evolving towards diversification. The connection strategy of the nodes in the deep neural network has attracted great attention from scholars recently, inspired by the exploration of natural phenomena. Event-triggered synchronized reaction-diffusion neural network was introduced into the research topic of deep neural networks (Song et al., 2023). To meet the challenge of finding an effective precursor of the failure and degradation regarding the complicated mechanism of mechanical fatigue, some novel methods (Feng et al., 2023a, 2023b) were proposed to develop a health indicator for the gearbox degradation assessment instead of labeling the RUL directly. The genetic algorithm (GA) was applied in this work to enhance the adaptive of the RNN-based approach automatically. Furthermore, a recent study has conducted a survey that demonstrates that the deep learning approaches have enhanced the predictive maintenance that was enabled by digital twin (van Dinter et al., 2022). They indicated that the digital twin platform will be an effective solution for replacing high-cost aging accelerating experiments and improving the monitoring of system parameters, as a result, providing more high-quality training data and improving the performance of predictive maintenance. Another significant innovation related to the neural network is a self-attention mechanism, which integrates a self-adaptive feature extraction mechanism for the neural network architecture (Xiao et al., 2022). These advancements are developed based on the enhancement of traditional deep neural networks, to meet the practical requirements of the industry applications. To explain the current situation of the development of deep learning architecture in predictive maintenance, the review of the evolutionary trajectory of deep learning will be described in the following section.

This paper reviews the development of predictive maintenance under Industry 4.0 and the application of deep learning approaches in industrial predictive maintenance. As a matter of fact, a substantial number of recent publications have conducted the review of deep learning or artificial intelligence techniques for the prognosis and fault identification regarding the challenge of reliability development of practical industrial applications (Fink et al., 2020; Maher and Danouj, 2020; Qiu et al., 2023). Typically, they tended to analyze the overall development of deep learning-based industrial applications in terms of the popular advancement assisted by the DL techniques, such as image processing, decision-making, and pattern recognition. According to a portion of the literature we have explored, it's common for researchers and engineers to integrate several types of architecture considering the situation where multiple kinds of parameters should be parsed. At the same time, most of the current studies did not indicate the advance of specific architecture in terms of the inimitable feature of input data. Thus, there are still some blanks that should be identified by a comprehensive survey. Most of the existing works did not make a contribution to compare the limitations and advantages of deep learning architecture based on the techniques analytic of input signal and data collected from practical usage, especially regarding the practical predictive maintenance problems in various industry scenarios. It's meaningful to provide comprehensive guidance in selecting appropriate architecture for encountered input signals or data types in the context of confronting one or multiple types of data collected from industrial devices. Therefore, the contribution of this paper could be organized as.

- To provide a brief prior understanding of engineering applications, various studies, applications, methods, and predictive targets within the scope of predictive maintenance are summarized in the following sections. As such, this study seeks to identify clear and decisive factors to aid researchers and practitioners in correctly selecting deep learning architectures for predictive maintenance.
- We surveyed 248 peer-reviewed papers published between January 2017 and April 2023 and selected 106 as the primary studies The strategy of setting up the keyword for the search engine is identified to improve the accuracy of addressing the mentioned challenges in industries. In order to substantiate the essential of applying suggested deep learning architecture to the corresponding task, the filled gap with respect to the reason collected from each article is indicated.
- Our findings suggest that each deep learning architecture has its application scope, advantages, and limitations for predictive maintenance. This is summarized through an in-depth analysis of each deep learning architecture with supporting guidelines for selecting an optimal deep learning architecture for a particular predictive maintenance application.

The rest of the paper is organized as follows. Section 2 introduces predictive maintenance. Section 3 overviews deep learning-driven predictive maintenance. Section 4 describes our survey method. Section 5 presents the survey results, including the five deep learning architectures with their advantages, limitations, and application domains. Following this, supporting guidelines and decision factors for selecting deep learning architectures are discussed in Section 6, together with a comparison to similar research papers. Section 7 concludes the paper and suggests possible future research.

2. Predictive maintenance

According to the European standard EN 13306:2010, maintenance is defined as the "combination of all technical, administrative and managerial actions during the life cycle of an item intended to retain it in or restore it to a state, in which it can perform the required function" (CEN/TC319, 2010). Further, maintenance can be divided into three main categories: corrective, preventive, and predictive maintenance, which have been detailed in (Wang et al., 2015). In EN 13306:2010, predictive maintenance is defined as "condition-based maintenance carried out following a forecast derived from repeated analysis or known characteristics and evaluation of the significant parameters of the degradation of the item." Traditionally, condition-based maintenance

schedules actions based on information about the *asset's current condition*. In contrast, predictive maintenance schedules actions based on forecasts from analytical models of the estimated *future* condition, degradation development, and RUL of the asset. With the growing demand for resource efficiency and productivity, executing maintenance at the right and optimal time is increasingly vital. Through Industry 4.0, the increasing availability of real-time data has allowed manufacturing companies to explore alternative maintenance methods that no longer need to follow predetermined and rigid maintenance schemes, often inhibiting product flow. Maintenance can be performed only after analytical models predict that enough indicators are present to predict an asset's degradation or impending failure accurately.

As an advanced and ideal maintenance strategy, predictive maintenance includes all activities carried out to maximize the service life of target equipment under the promise of zero failures by assessing the real states and predicting the lifetime of the equipment. Through intelligent analysis of (big) data from condition monitoring and operation, predictive maintenance can reduce costs by reducing the number of unnecessary scheduled time-based maintenance operations. The main functions can be conducted by collecting data on the equipment under operation and discovering potential faults or degradation through the historical analysis of similar equipment and knowledge acquired over time. When combined with AI techniques, which work similarly to a human brain, we can further identify patterns and information within the data of potential failures.

Studies show that some of the main challenges of predictive maintenance are how to extract features effectively and appropriately from collected data (Deutsch and He, 2016), how to identify and classify failures flawlessly (Qu et al., 2018), and how to predict the potential faults or degradation precisely (Zhang et al., 2017). Many effective and useful methods have been presented, researched, and applied in feature extraction or signal analysis in predictive maintenance. However, since feature extraction is highly subjective, the optimal method usually depends on the actual case and problem at hand. Besides, establishing a proper failure identification or prediction system requires highly skilled engineers and considerable domain knowledge to extract valuable and significant information. As a growing trend in data analysis, deep learning, with its multiple-feature support, may offer an alternative method for automatic data interpretation rather than the manual interpretation of data by human experts.

3. Overview of deep learning–driven approaches for predictive maintenance

3.1. Development of deep learning

Deep learning focuses on constructing deep hierarchical models to enable a machine to "learn" from raw data (LeCun et al., 2015). Deep learning hypothesizes that a hierarchy of intermediate representations is required to learn high-level data representations. In contrast, shallow learning's archetypical learning model can simply be a feedforward neural network with one input, output, and hidden layer. However, initial deep learning methods were developed from these shallow learning models, as was the case with artificial neural networks based on the backpropagation learning method. However, in the late 1980s, researchers started to realize that it was challenging in practice to apply backpropagation-based training methods to deep neural networks (DNNs) with multiple hidden layers (Rumelhart et al., 1986). In 1991, research by (Hochreiter, 1991), which has been considered a milestone of explicit deep learning, formally identified the primary cause of this difficulty; that is, typical DNNs have the problem of gradients vanishing or exploding. Standard activation functions cause accumulative differences from backpropagation to either decay or explode exponentially through the number of layers.

Since then, several approaches have been proposed to mitigate this fundamental problem. Today, these approaches have evolved into different deep learning architectures. As (Deng, 2012) reports, "deep learning refers to a rather wide class of machine learning techniques and architectures, with the hallmark of using many layers of non-linear information processing stages that are hierarchical in nature." Such an architecture enables better accuracy and performance when processing large and complex data sets.

3.2. Application of deep learning in predictive maintenance

In line with the motivation behind Industry 4.0, the rapid expansion in the scale and application of modern manufacturing have significantly increased the inherent complexity and diversity of industrial equipment and manufacturing systems. This trend has made it difficult, and in most cases almost impossible, to predict or recognize potential faults and critical equipment conditions efficiently and timely using only traditional means such as statistical analysis.

Machine learning–based techniques stimulated the development in many fields, from sentiment analysis to spam filtering and information retrieval to AlphaGo (Silver et al., 2016). Nevertheless, directly applying machine learning–based technologies in predictive maintenance is challenging. As highlighted by (LeCun et al., 2015), "Conventional machine learning techniques were usually limited by their ability to process natural data in their raw form." In addition, forecasting potential failures in the future is much more challenging than ordinary diagnostics due to the apparent absence of working condition data of the equipment or system in the future. Thus, high-complexity data-driven models are necessary to identify and track the progress of potential failures, which is an inherently challenging task due to the requirement of extracting highly abstract features from various attributes from raw data. In recent years, research on deep learning methods has largely stimulated their applications in predictive maintenance (Gan and Wang, 2016) because "the family of deep learning approaches has been growing increasingly richer, encompassing a variety of neural networks with multiple processing layers, hierarchical probabilistic models, and different kinds of unsupervised or supervised feature learning algorithms" (Li et al., 2017). Deep learning may discover and capture useful and discriminative information automatically in a self-learning manner from industry data for predictive maintenance. To date, various types of deep learning architectures, such as stacked autoencoders (SAEs) (Kumar et al., 2017), deep belief networks (DBNs) (Li et al., 2017), and fully connected DNNs (Wang et al., 2017), have been applied successfully in predictive maintenance.

Nevertheless, a general problem that needs to be understood ahead of suggesting any deep learning architecture in practical applications is investigating the influence of industrial datasets on the feasibility of developing deep learning methods. In this study, the datasets of the reviewed papers are indicated in the following content, of which the majority were collected from the laboratory and practical industrial. Owing to the principle of supervised learning, most of the predictive maintenance requires a training dataset that contains the observation of degradation, failure, and even a complete run-to-fail dataset. However, the mechanism of degradation and failure of components and equipment are extensively complicated especially when the analysis of the operation condition was involved in the health assessment. Eventually, some studies have pointed out that the dynamic behavior of the precursor of the health condition varies in the aging experiment(Xiao et al., 2022). In this context, new requirements for the acquisition of training datasets should be incorporated into the experiment setup and case study to obtain representative datasets for deep learning development. Typically, at the phase of designing the development, researchers need a complete awareness of the mechanism of degradation and failure so that the trade-off option can be considered to meet the requirement in the practical applications. Namely, applying an appropriate arrangement of the sensor to the experiment platform is fundamental to establish a representative effective training dataset. The density, and duration of the sensing characterize the content of the training data, which could be

learned by the deep learning model. On the other hand, according to the signal processing requirement, such as the Nyquist theorem, the sampling frequency of the sensing must be fulfilled based on the hidden pattern analytic. The robustness and feasibility of the experiment design impacts the fidelity of the training dataset from the capability of simulation, in terms of the practical operation condition. The experiment design usually includes several incremental stages such as hardware platform implementation, failure injection, and data collection. The key factors of the deterioration, such as extrinsic or intrinsic pressure and vibration, overcurrent, and overheating, should be strictly controlled by the control system with which the aging experiment platform is accompanied. Furthermore, the concept of the prognosis seeks to establish a relationship between the parameter measurement with the degradation and failure pattern, where some novel papers have explored further to develop online prognosis(Zhang and Li, 2021; Zhang et al., 2021). In the worst case, the representative data of the health condition may not reflect the actual failure pattern of the objective, but only the boundary condition. However, owing to the novel development of online learning, the collection of deterioration data becomes unnecessary based on the categorization of the failure boundary analytics (Khan and Yairi, 2018). In the most of cases studied by the reviewed papers, the training dataset usually contains the measurement of significant parameters, along with several auxiliary parameters measurements to maintain the reliability of the prediction result. Especially, the measurement should reflect a comprehensive condition of the measured system aligned with the timestamp. Therefore, more discussion will be provided in the following section after the demonstration of the survey results.

4. Research method

4.1. Research questions

This research aims to summarize the applied deep learning architectures for predictive maintenance and provide supporting guidelines for researchers and practitioners to select an appropriate deep learning architecture based on their predictive maintenance context. In this section, each architecture is evaluated based on their performance in the experiments that carried out by the authors. Fundamentally, mean square error and accuracy are the metrics commonly used in the reviewed papers, which explains the reason for applying the suggested method based on the benchmark comparison. Specifically, for the regression task such as RUL prediction, the MSE and the coefficient of determination were used to evaluate the performance. For classification tasks like fault identification, accuracy was the first choice. Especially, owing to the training strategy of SAE, reconstruction loss may also be considered to evaluate the performance.

We have tried to answer the following research questions.

RQ1. What are the main focuses of each deep learning architecture in predictive maintenance?
RQ2. What are the reported advantages of each deep learning architecture?
RQ3. What are the reported limitations and challenges of each deep learning architecture?

4.2. Searching and selection of primary studies

This study aims to provide guidelines for selecting deep learning architectures and summarize the main applications of deep learning architectures in predictive maintenance in the industry. The academic database of Scopus was selected for its broad coverage and relevance. The search string "("deep learning" OR "deep neural network") AND (maintenance OR failure OR fault OR degradation) AND (prediction OR forecast OR assessment) AND (industrial OR industrial data)" resulted in 248 peer-reviewed articles published from 1st January 2017 to 16th April 2023, under the subject area of computer science and engineering. The number of papers published on this topic has increased drastically in recent years, as shown in Fig. 1. The low number of articles in 2023 is because our search was finished on the 16th of April 2023.

Firstly, we reviewed the title, keywords, and abstract of all 248 articles. A total of 72 papers were found to be outside of our study scope. For example, real-time traffic forecasting (Pérez et al., 2018) and quadrotor obstacle avoidance (Yang et al., 2019b) were excluded. We read the full text of the remaining 176 papers, focusing on applying deep learning specifically in predictive maintenance. We excluded papers that focused only on newly developed methods without extensive use, such as hysteresis loop analysis-based deep learning networks (Zhou et al., 2019), due to their lack of validation by other researchers working on this topic. We also excluded papers reporting on *integrated structures* (Deutsch et al., 2017b; Listou Ellefsen et al., 2019) because of the lack of explanation, or comparison, about selecting a deep learning structure. Lastly, papers of which we do not have full-text versions (*not available*) and a paper that is withdrawn (*retracted*) were also excluded. An overview of the 248 papers' inclusion and exclusion is given in Fig. 2.

In total, 106 papers, that interpret the advantages and limitations of deep learning methods for predictive maintenance, were chosen as primary studies and analyzed in depth to answer the RQs. Based on the thematic analysis, we categorized the main deep learning architectures for predictive maintenance into five categories: DNNs, SAEs, DBNs, deep recurrent neural networks (DRNNs), and convolutional neural networks (CNNs). The following section explains the results of analyzing the papers applying the five deep learning architectures in predictive maintenance to answer the RQs.

5. Deep learning architectures in predictive maintenance: research results

5.1. Deep neural networks

A DNN with backpropagation evolved a traditional feedforward neural network, stacked by several levels of perceptrons, which are fully connected by each other. Usually, the capability of learning nonlinear or other complex patterns is trained by a backpropagation algorithm (Bhakte et al., 2022). This type of network is one of the original deep learning approaches. Although generic types of deep learning networks can be called DNN, we distinctly mark networks with fully connected hidden layers and backpropagation training as DNN to distinguish them from other types of deep learning architectures. Initially, researchers found it difficult to train a multilayer network with several hidden layers

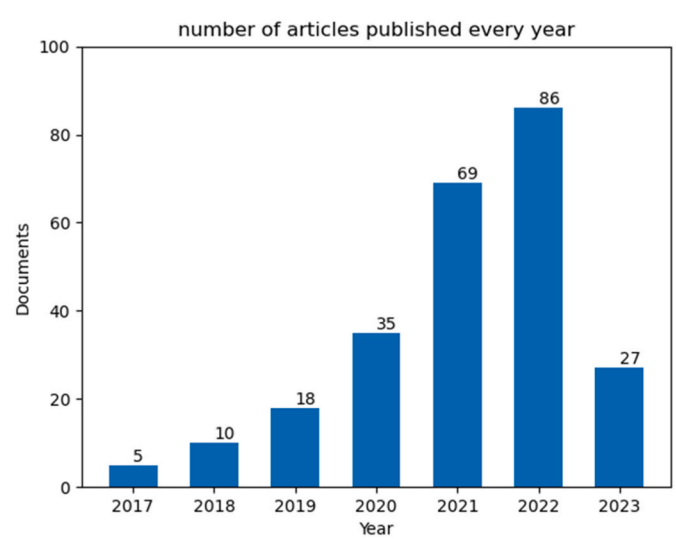

Fig. 1. Number of papers published from January 2017 to 16th April 2023.

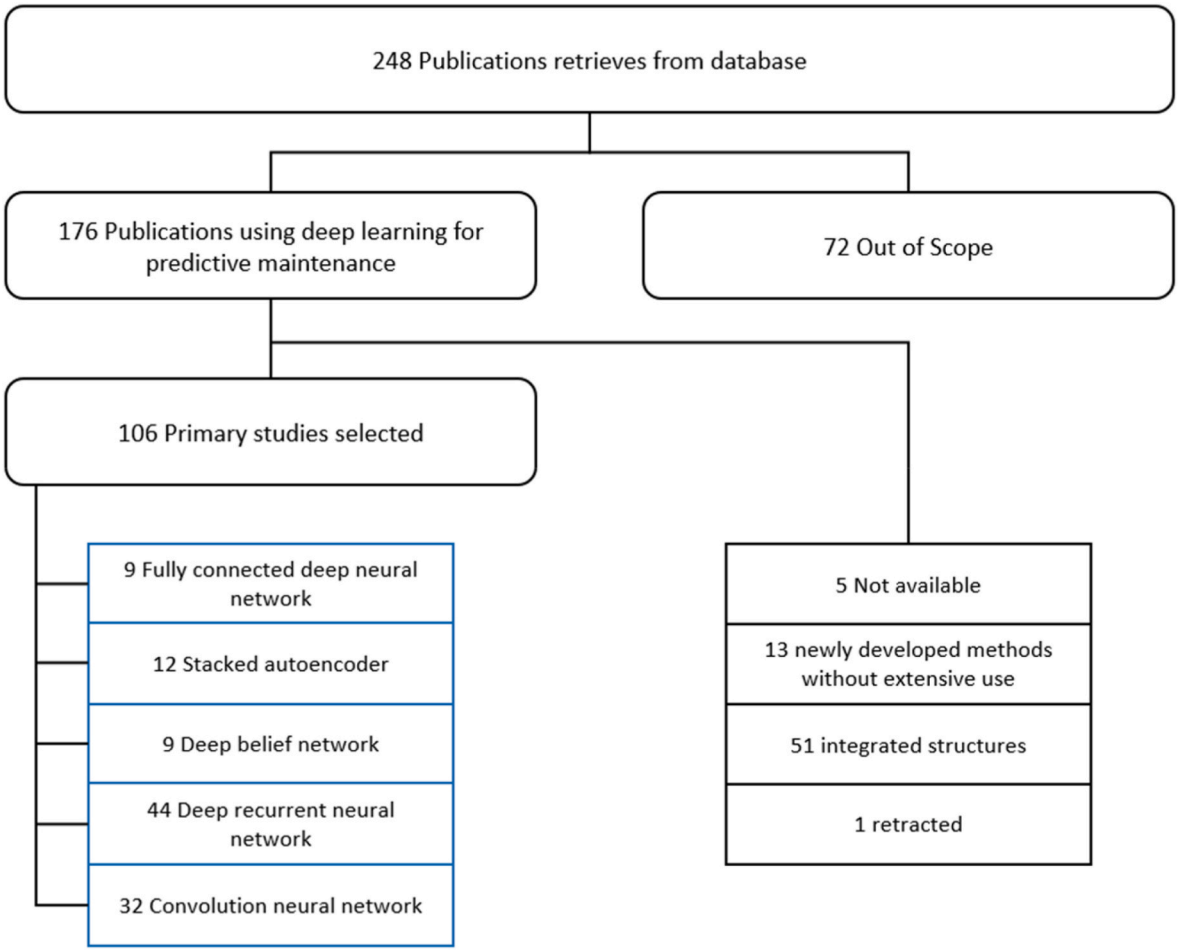

Fig. 2. The selection and filtering of primary studies.

with real-life data. However, the advancement of available computational power and the development of training algorithms have reduced propagating errors and made it feasible to train multiple layers. Besides, modern DNNs usually combine more advanced functions such as a rectified linear unit (ReLU) or Maxout and a more efficient learning process within their deep structure, including parallelizing and momentum-based gradient descent, which makes DNNs successfully applied to achieve various targets for predictive maintenance (PdM).

5.1.1. *Result of RQ1: main tasks of DNNs*

Table 1 shows the following task-oriented classification.

- RUL prediction (2 papers)
- failure prediction (1 paper)
- failure identification (4 papers)
- degradation assessment (2 papers)

According to the papers mentioned above, the most frequently applied task of DNN is failure identification, followed by RUL prediction and degradation assessment (Bhakte et al., 2022). designed a 4-layer deep ANN for two types of failure classification, where accuracy reached more than 99 percent. The information from different types of sensors was fused within the first hidden layer of the DNN. The interesting part of their result is the gradient distribution illustration regarding the supervised output vector. The result indicated that DNN presents the capability of recognizing the relative importance of the input and output variables, which is meaningful to the interpretability of black-box models (Li et al., 2019e; Sandhu et al., 2023). applied and evaluated the performance of deep ANN based on the back-propagation algorithm in the degradation assessment that mapped the vibration data feature to degradation locations and stages, respectively. Owing to the complexity of high-frequency data, a deeper architecture was developed in this study, which also provided a promising result. The activation function enabled the ability to fit non-linear functions at the hidden layer followed by them. In most practical implementations, the activation function significantly formed the capability of the DNN (Ren et al., 2017). designed a DNN with eight layers that aim to predict the percentage of RUL of multi-bearing collaboratively, coupling with "ReLU" and "Sigmoid" activation functions resulting in better accuracy than most of the major machine learning methods.

5.1.2. *Result of RQ2: advantages of DNNs*

The common advantage mentioned by these references is the capability of fitting complex system functions between the industrial equipment operation condition data and the failure or RUL (Bhakte et al., 2022). For example, deep fully connected ANN is developed from a conventional feedforward 3-layer multi-layer perceptron, enhanced by deeper architecture and BP algorithm. Reliable performance could be obtained by applying the DNN based on the appropriate setup of parameter optimization. Therefore, the structure of DNNs is much more flexible and interpretable, compared to other complicated DL methods, whilst it presents a much stronger ability to overcome overfitting and underfitting problems by combination design of multi-layer perceptron and activation functions (Ren et al., 2017; Li et al., 2019e) found that DNNs perform well at mapping degradation and identifying potential failures in a complex environment. Accordingly, the relatively simple structure of DNN provides researchers with a trade-off option to consider an efficient method with a more flexible structure and low

Table 1
Applications of DNNs in predictive maintenance.

Ref.	Task	Output	Filled Gap	Advantages or reasons for using the structure	Input signals or data type	Other methods with which this method was compared
Ren et al. (2017)	RUL prediction	RUL of a multi-bearing	Improve the accuracy from model-based and knowledge-based methods; Address the issue of collaborative analysis upon a group of rolling bearings in similar working conditions.	The computation and storage complexity of a DNN is lower than other architectures while achieving a satisfactory performance	25.6 kHz bearing vibration signal	Gradient-boost decision tree, SVM, Gaussian regression, Bayesian Ridge, 3-layer BPNN
Li et al. (2019e)	Degradation assessment	Health condition of the spindle	Concerning the complicated pattern of deterioration with the life cycle of mechanical equipment. The conventional quantitative analysis presents an unsatisfactory performance that can't meet the requirement of accurate assessment.	A DNN performs well in mapping degradation and identifying potential failures in a complex environment	Vibration signals	BPNN, SVM, K-NN, DBN
Kakavandi and Larsen (2022)	Fault identification	Fault class and relevance map	Traditional machine learning methods can't interpret the MLP-based black-box model to figure out the most critical feature that leads to the product assembly failure	Full connected DNN, or multi-layer perceptron, is usually applied to fit most nonlinear and complex systems.	Torque signal of the assembly robot	Not mentioned
Du et al. (2023)	RUL prediction	RUL	Some deep learning methods, such as RNN and CNN, rely on long periods of training and fine-tuning. Correspondingly, complex architectures might be sensitive to noise signals, which leads to inaccurate RUL prediction.	A full-connected neural network-based transformer reduces edge devices' computational and storage complexity while maintaining satisfactory accuracy.	C-MAPSS	Training time: LSTM-CNN and GCU-based Transformer RMSE: CNN, RNN, and various transformers based on them
Sandhu et al. (2023)	Degradation assessment	Degradation location and severity	The behavior of nuclear piping is more complex than civil structures, while the previous studies only focus on a single damage-sensitive feature, and they can't realize minor damage and uncertainty	Once the architecture and parameters of a deep ANN are designed well, the prediction can achieve high accuracy depending on the utilization of the back-propagation algorithm	Measurement from the sensors mounted on the pipe at different locations	Not mentioned
Bhakte et al. (2022)	Fault identification	Fault type	Some deep learning methods are not easily interpreted, which places low trust. Thus, it's a challenge to deploy some deep learning methods for the operator who has responsibility for safety.	The flexible architecture of MLP extended the adaptive capability when dealing with potentially diverse input and output data formats.	Chemical process parameters such as water flow and temperature	Not mentioned
Filz et al. (2021)	Fault identification	Binary classification task	Traditional preventive management has shortcomings that are mainly about uncertainties and efficiency. Also, there is a research gap between historical data and data-driven decision support that can be provided dynamically for high-value industrial equipment.	The recall of deep NN is higher than other traditional machine learning methods based on experiments.	Time series that denotes the normal or failure scenario.	Naive Bayes inference, Linear and logistic Regression, decision tree (DT), Gradient Boosted Trees, Random Forest (RF).
Kafunah et al. (2021)	Fault identification	Type of fault	Robust feature extraction and data augmentation are the majority selection for imbalance dataset pattern recognition, as a result, most of the existing data-driven methods can't adapt to imbalance datasets.	"The architecture of DNN is proved to be able to counter imbalanced dataset by pre-training using a larger dataset as the source domain and eventually transferring the knowledge to target domain."	Pressure, vibration signal	GPR-based GAN, relative likelihood approach
Zeng et al. (2022)	Fault prediction	Failure label	DNN may present a robust performance of learning the hidden pattern from a dataset through a robust architecture. While the architecture may be set by various weight combinations.	Deep neural networks enhanced the traditional neural network with good feature extraction ability.	Time series data of commercial air conditioners	Not mentioned

computation and storage complexity.

Other advantages of DNNs are also mentioned in the studies, such as their ability to allow for anomaly classification without prior knowledge (Siegel et al., 2018), to map probabilities of multiple health stages, and to indicate degradation processes (Xia et al., 2017) and its capability in big numeric data analysis (Chen et al., 2018).

5.1.3. Result of RQ3: limitations of DNNs

The design of the DNN structure needs careful consideration of the number of neural at each hidden layer and other key parameters, huge among of historical data that covered target condition (Ren et al., 2017; Sandhu et al., 2023), as well as a reasonable training strategy. Otherwise, the accuracy may be unsatisfactory even after a long training process. On the other hand, the input of DNN should be representative features according to all references above, so a reasonable feature extraction technique should be carefully considered before applying DNN (Li et al., 2019e). had proved that the selection of feature extraction also depends on expert knowledge in most scenarios, such as Wavelet package decomposition (WPD), Ensemble empirical mode decomposition (EEMD), etc. Specifically, as mentioned above, the

classification process of DNN is much more explicitly explainable than other deep learning models. However, it's still much harder to explain than conventional machine learning methods (Bhakte et al., 2022).

5.2. Stacked autoencoders

Another feasible solution to vanishing gradients is using unsupervised pretraining and constructing hierarchy networks. The idea is to make each hidden layer predict the input of the next layer individually in an unsupervised manner. Based on this idea, two essential types of neural networks for fault diagnosis and prognosis, SAEs and DBNs, were invented.

SAEs are constructed by stacking multiple autoencoder layers. The architecture is usually trained by the input data in each layer to find the most suitable manner of encoding, which can help to extract nonlinear features without labels. Thus, an SAE is also a general method for dimensionality reduction. All hidden layers will first be trained unsupervised to learn multiple nonlinear transformations from the training dataset, which may assist subsequent models to study the essential variations and find the discriminative information from raw sensory signals. Thus, SAEs have been widely and successfully applied in dimensionality reduction and other aspects of predictive maintenance, as shown in Table 2.

5.2.1. Result of RQ1: main tasks of SAEs

In the 12 primary studies identified that applied SAE architectures in predictive maintenance, the main tasks include.

- failure identification (5 papers)
- degradation assessment (1 paper)
- RUL prediction (4 papers)
- failure prediction (1 paper)

The structure of SAE is divided into the encoder and decoder. Owing to the ability of low-dimension latent representation learning and unsupervised learning, SAE is widely utilized in feature extraction for industrial signals (Ren et al., 2018; Xia et al., 2019). indicated that the deep SAE played a role in time-domain signal feature compression and extraction technique. The abstract latent feature could be obtained from the latent layer by minimizing the reconstruction loss that defined by the developer. One of the popular options of the reconstruct loss is MSE, meanwhile, Kullback–Leibler(KL) divergency incorporates the probability distribution of the input data into the training process. The authors successfully utilized the SAE to drop redundant time-domain features and performed better than traditional feature extraction techniques. To address the drawback of not being able to "consider the local structure-related information," (Yang and Wang, 2023) introduced the structure of SAE as a hierarchical structure that couples with multiple AE, which are connected by each other (Xu et al., 2022). proposed a more flexible SAE structure that embeds the battery's capacity at the current cycle to the middle-level layer to predict the capacity at the next working cycle in the latent layer of SAE. Hence, the operation condition could be obtained from the decoder and verified by physical equations. As (Maurya et al., 2021) described: "To improve the learning power of the network, multiple sparse AEs are stacked, which is known as a stacked sparse AE (SSAE)." They stated that the challenge of traditional methods is the time-varying harmonics and inclusion of random signal components, which fit the complicated feature extraction ability of SAE. Specifically (Ribeiro et al., 2022), stated the research challenge is that most industrial data are collected from the normal condition, leading to unbalanced training process issues, thus in this context, the utilization of SAE "learns normality rather than computing abnormality scores".

5.2.2. Result of RQ2: advantages of SAEs

The common advantage mentioned by our results is the promising capability of learning significant features from input vectors in unsupervised learning. It's worth noticing that (Ribeiro et al., 2022; Tang et al., 2021) mentioned SAE-based solution of fault and anomaly detection can "automatically identify the feature in the data that can be used in classification algorithm" and "be adapted to dynamic environments when there are frequent data updates." Apparently, modern SAE has evolved adaptively efficient feature engineering capability. Additionally, regarding the nature of SAE's hierarchical structure (Yang and Wang, 2023), introduced the idea of a pre-trained and fine-tuning strategy to extract non-local, local, and global features. The nature of SAE presents the ability to promise a robust performance in constructing efficient feature space and reconstructing the input space. They figured out that the SAE can "preserve the locality manifold structure of data via establishing a clustering layer."

Thus, we consider the main advantage of SAEs is their ability to extract hierarchical representation features from complex raw data automatically and adaptively, meanwhile, to discovering discriminative information about the failures. Under the circumstance of involving multiple features collected from different sensors, the performance of SAE, which was used to predict the RUL, was better than DNN.

5.2.3. Result of RQ3: limitations of SAEs

The main limitation of SAE is the requirement of the reasonable manual design of latent space between the encoder and decoder, which represents the number of useful features of the input vector. Most reviewed papers in this category successfully identified the current condition of the target equipment or system. Predicting the condition of failures in the future may require other methods. For example (Xia et al., 2019), first applied an SAE to extract health-related indicators of bearings and then predicted the RUL through a shallow neural network (Ren et al., 2018). used BPNN to obtain the final RUL of bearing after using SAE to learn hidden patterns from vibration signals (Yang and Wang, 2023). indicated the drawback of SAE that "The selection of hyperparameters requires specific analysis" and "Due to adding the nonlocal, local and global regularized objective to SAE, the training time cost is high."

Thus, we consider that it is hard to use an SAE alone to predict failure or RUL. Also, the good performance of SAE relies on careful consideration.

5.3. Deep belief networks

A DBN, proposed as a novel probabilistic-based network architecture by Hinton et al. (2006), is another type of deep learning architecture that utilizes the pretraining of hierarchy networks. This type of network is constructed by learning several hidden layers of restricted Boltzmann machines (RBMs), which apply an energy function to describe the probability of each possible pair of visible and hidden vectors. In stage k, the output in the previous layer $(k-1)^{th}$ is used to train the kth layer (concerning an unsupervised criterion), while the previous layers are fixed.

In a DBN, every RBM layer learns to encode a probability distribution through the weight matrix that indicates actions in the visible layer from the hidden layer. A DBN can extract representative information from the input data by minimizing the energy in the constructed network. Our review results show that this energy-based model can capture intrinsic characteristics such as capacity degradation (Zhu et al., 2022a) and fault features of bearings (Tang Hoang and Jun Kang, 2018) for predictive maintenance. More detailed information from the surveyed papers is shown in Table 3.

5.3.1. Result of RQ1: main tasks of DBNs

We found nine papers that successfully applied DBN architectures in predictive maintenance. The main tasks of them are as follows.

- Fault identification (1 paper)
- Degradation assessment (1 paper)

Table 2
Applications of SAEs in predictive maintenance.

Ref.	Task	Output	Filled Gap	Advantages or reasons for using the structure	Input signals or data type	Other methods with which this method was compared
Ren et al. (2018)	RUL prediction	Time-domain feature of vibration signal	Effective and robust deep learning-based approaches for bearing RUL prediction still need more exploration, which has also been proven to recognize a much more accurate RUL estimation at the early stage.	The SAE method can extract critical information from the vibration signal without increasing the scale of stacked layers and presents a better accuracy than traditional machine learning methods.	Raw vibration signal	None-autoencoder BPNN, BONN with FSPS, SVM
Xia et al. (2019)	RUL prediction	Degradation stages of the bearing	Prior expert knowledge affects the effectiveness of signal feature extraction significantly, which may refer to the most associated linear or nonlinear relation to the health state of different types of bearings that work on various loading conditions.	Instead of applying specific feature extraction approaches manually, SAE can identify the health state from the raw measurement for the monitored bearing.	Vibration signal	Not mentioned
Yang and Wang (2023)	Fault identification	Normal or abnormal identification based on the threshold comparison	High-qualified process monitoring needs to analyze the latent features of raw data and project them into subspace. As a result of typical deep learning methods such as DNN, origin AE, DBN, and CNN usually ignoring local structure-related information, they are too sensitive to noise.	The proposed SAE can transform the raw data into an efficient feature space considering global and local manifold structures.	Industry process sensor data.	PCA
Xu et al. (2022)	Degradation assessment	The health condition of Lithium-ion batteries	Conventional SOH prediction methods use discharge capacity, which is insufficient for accurate degradation assessment and may need much prior knowledge. Another gap is that some deep learning methods, such as CNN and SVM, tend to present an overfitting or underfitting problem. Also, interpretability is the challenge of data-driven methods.	SAE can replace the traditional specific feature engineering with an interpretable network structure, whilst presenting a satisfactory performance for the estimation of the health state.	The capacity, temperature, and terminal voltage at each working cycle.	Gaussian process regressor
Maurya et al. (2021)	Fault prediction	Fault type	Most of the existing health monitor applications apply traditional signal processing methods to extract features fed to deep learning models. They are facing challenges, for example, random high-frequency harmonic components and noise signals.	Owing to the capability of feature mining from multi-dimension time series, SAE presents a satisfied nonlinear pattern recognition ability with fewer sensor measurements.	Real-time sensor data of air compressor.	SVM, RF, Softmax classifier
Pillai and Vadakkepat (2021)	RUL prediction	Normalized health index	The result of traditional feature engineering was usually affected by noise, outliers, and unexpected information, which directly affect the final accuracy of the deep learning classifier significantly.	Stacked AE enhances the automatic ability of low-noise feature extraction from traditional AE, which learns a low-dimension robust feature vector from a non-stationary time series.	Engine state time series from sensor data.	RULCLIPPER, MODBNE, CNN, DCNN, CNN-XGB, LSTM, RBM-LSTM,
Tang et al. (2021)	Fault identification	Fault type of bearing	Conventional ANN can't extract satisfied features from signal and present over-fitting and under-fitting problems. Many unlabeled data are produced in the industrial system, and labeling costs too much time and funding.	SAE is used to identify the feature automatically and solve the unbalanced dataset problem.	Bearing dataset	Pre-trained DNN
Jakubowski et al. (2022)	Fault identification	Normal or anomaly label	Cluster-based anomaly detection must set up parameters depending on the number of "neighbors" or the radius of the normal group.	The architecture of SAE is flexible. The compression of information at the latent layer is useful for fault detection since it represents the most related feature.	C-MAPSS	VAE
Guo et al. (2023)	RUL prediction	Nonnormalized RUL value	The structure of industrial equipment becomes more and more complicated, so the physical model is hard to	Self-attention mechanism-based SAE can extract features more efficiently and eliminate the vanishing gradient difficulties.	C-MAPSS	

(*continued on next page*)

Table 2 (*continued*)

Ref.	Task	Output	Filled Gap	Advantages or reasons for using the structure	Input signals or data type	Other methods with which this method was compared
			describe by traditional mathematical methods.			
Ribeiro et al. (2022)	Fault identification	Fault indicator	Most of the existing data of the screw-tightening process are normal examples. However, in the context of only training on normal data, high-accuracy anomaly detection is required based on a data-driven model.	The classification from SAE can retain high accuracy adaptively when the assembly process change		Random forest, isolation tree
Silva and Droguett (2021)	Fault identification and RUL prediction	Fault mode and the number of cycles until failure	Traditional machine learning methods may be challenging to achieve satisfactory accuracy when the dimension of input data is high. Also, they need a large amount of labeled data that could be hard to be collected/	SAE could be trained upon an unlabeled dataset; also, it transforms high-dimension data to low-dimension latent representation.	CMAPSS,	Not mentioned

- RUL prediction (3 papers)
- Failure prediction (4 papers)

According to our review, the main tasks of applying DBN architectures are RUL prediction and failure prediction. The most mentioned challenge that DBNs address is conventional complicated feature engineering upon comprehensive nonlinear systems. For example (Tang Hoang and Jun Kang, 2018), indicated the main research challenge is that in the case of multi-sensor fault diagnosis, disordered sensor measurements for a system were found difficult to fuse directly, and then transform into validity feature information (Deutsch and He, 2018). mentioned that traditional methods require complex signal processing technologies for feature extraction or prior knowledge to establish explicit models, which is infeasible in practical applications, meanwhile (Deutsch et al., 2017a) also stated that most current prognosis strategies still largely rely on expert knowledge for establishing mathematical model and analyzing complex measurement. Specifically (Zhu et al., 2022b), not only indicated the traditional data-driven methods' dependency on feature engineering but also raised their concerns about the limited information revealed by most popular feature extraction approaches, as a result, without an interpretation of corresponding confidence intervals.

5.3.2. Result of RQ2: advantages of DBNs

The most common advantage of DBNs mentioned by our result of the review is the specifically robust prediction performance on the complex probabilistic pattern recognition, depending on the hidden layer training. For example (Haris et al., 2021), proved the extensive learning capability of DBN's multi-layer architecture in the capacitor's RUL estimation, assisted by Bayesian optimization. Their result illustrated excellent performance by training on a dataset collected from a noisy working environment for a supercapacitor degradation, in which training data only require less than 6% of whole life-cycle data. On the other hand (Zhu et al., 2022a, 2022b), pointed out the probabilistic-based network structure explores another way to address the challenge of tracking the posterior distribution of high-dimension data when using a sophisticated DNN architecture. This means that another advantage of DBN is incorporating the confidence interval of the training samples (Deutsch and He, 2018). concluded that the DBN-based deep learning model can self-taught feature learning in practice due to its unsupervised pre-train strategy, in other words, it can recognize the health state from both raw measurements like vibration signals and feature information extracted from the former.

From the abovementioned content, it has been clearly clarified that DBN can guarantee the robust ability of high-dimensional probabilistic pattern recognition.

5.3.3. Result of RQ3: limitations of DBNs

Like the deep learning methods mentioned above, one of the limitations is the selection and determination of hyper-parameter of DBN, requiring careful consideration (Haris et al., 2021). Likewise (Tang Hoang and Jun Kang, 2018), raised their concern about seeking an effective paradigm for setting up and fine-tuning hyper-parameters manually for deep learning models like DBN, to adapt to various circumstances (Deutsch and He, 2018). utilized grid search to identify the best hyperparameter of DBN (Akpudo and Hur, 2020). indicated that overfitting, computation complexity and interpretability problems are concerned. What's more, the feasibility of DBN is established based on big data and the hyper-parameter issue, which are still a bottleneck for industrial applications. Specifically, the authors also pointed out that the DBN interpretation relies on expert statistical knowledge.

5.4. Deep recurrent neural networks

(Hochreiter and Schmidhuber, 1997) state that a novel deep architecture was developing, currently known as deep RNN (DRNN), to address the challenge of RNN, that insufficient and decaying error affected the training efficiency of traditional RNN. In recent years, researchers have proposed many DRNN variants, among which long short-term memory (LSTM) networks and gated recurrent unit (GRU) networks are the majority option and the most widely used. Speech recognition and language modeling, and most other sequence data processing and prediction are the successful research areas of DRNN-based applications. In DRNN, constant error carousels (CEC) are embedded as an activation function individually to establish a fixed weight mechanism, which guarantees that gradients will neither vanish nor explode during the computation of back-propagating through a CEC. By introducing a "memory cell," a DRNN network can keep previous states or information for calculating current and future steps. As stated by (Li et al., 2019d), for complicated scenarios, the analytics of information tend to depend on the utilization of multi-sensor measurement. Thus, instead of connecting the input and output with only traditional MLP, the feedback loop architecture assembled by one or multiple memory cells together with MLP, which includes forget gate and update gate, presents a much stronger capability of capturing nonlinear dynamics embedded in the multi-dimension time series. The networks were further used to study representations of the condition for predictive maintenance effectively. Table 4 lists ten primary literature on the applications of DRNN in predictive maintenance.

5.4.1. Result of RQ1: main tasks of DRNN networks

We found that 40 primary studies applied DRNN architectures in predictive maintenance and validated the performance of the established networks. The tasks that the studies focus on are.

Table 3
Applications of DBNs in predictive maintenance.

Ref.	Task	Output	Filled Gap	Advantages or reasons for using the structure	Input signals or data type	Other methods with which this method was compared
Deutsch et al. (2017a)	RUL prediction	Health index	Due to the high dependence on feature extraction, traditional data-driven prognostic methods rely heavily on signal processing and modeling based on expert knowledge.	DBN can learn the hidden pattern from a set of signal features. Meanwhile, it presents robustness capability and decreases the probability of reaching a local minimum	Vibration signal	Particle filter-based approach
Deutsch and He (2018)	RUL prediction	Scaled remaining work cycle	The traditional RUL estimation needs the complicated consideration of empirical feature engineering processing, or mathematical formulas that map the input measurement to the health state explicitly and manually. What's more, sometimes it's impossible to establish a mathematical model directly.	Owing to the nature of DBN, useful implicit information that represents the feature of measurement might be extracted, hence applied to learn the nonlinear relation.	Vibration signal	Particle filter-based approach
Tang Hoang and Jun Kang (2018)	Fault identification	Bearing health conditions	For a multi-sensor prognosis system, the fusion strategy is complicated so useful feature information is hard to parse out directly by explicit approaches.	DBN architecture can learn the multi-domain complex features from the measurement of vibration, which maps the health state through an implicit nonlinear relationship.	Vibration signal from electric motor under four load conditions	SAE, CNN
Akpudo and Hur (2020)	Failure prediction	Comparison of health index and failure threshold	Traditional machine learning methods require a lot of characteristics of samples. Feature extraction techniques like PCA depend on the amount of constituent hidden information and the structured nature of the fusion algorithm.	Unlike other deep learning methods, DBN illustrates versability in unsupervised learning and classification. It can learn the possibility distribution from the data.	Vibration signals	Gaussian process regression
Haris et al. (2021)	RUL prediction	State of health regarding working cycles	The RUL of supercapacitors is non-linear and has dependencies on some operation conditions. The measurement of supercapacitor degradation requires an aging experiment lasting a few years to figure out a precise result. Otherwise, it's hard to obtain a robust prediction.	DBN's multi-layer structure and BP-based learning strategy prove its extensive learning capability.	Working cycle	Not mentioned
Yuan et al. (2021)	Failure prediction	Regression curve of operation condition	Although deep learning techniques have demonstrated a strong capability of learning non-linear systems, most can't adapt to time-varying problems and need a long time of re-training.	DBN has proven capability of recognizing patterns from high-dimensional and large-scale industrial process data. A fine-tuned DBN could be updated rapidly and accurately online.	Flow, pressure, and temperature signal	Radial basis function network, and locally weighted partial least squares.
Chomacki et al. (2022)	Failure prediction	Damage classification	Increasing safety factors have led to multidimensional and undertaken problems, some of which are characterized by a large scale of uncertainty. As a result, computational method-based risk assessment may fail when the number of damage increases.	The bidirectional inferring structure of DBN fits diagnosis or prediction tasks	Horizontal measurement of soil deformation.	Probabilistic neural network, SVM, naïve Bayes classification
Zhu et al. (2022a)	RUL prediction	Life percentage	Existing machine and deep learning methods require careful development under passive learning. Obtaining massive industrial data usually consumes huge computation resources and funding.	The prediction of RUL could be obtained in the case of unsupervised learning on small-scale datasets.	Ball bearing vibration signal; Voltage for fixed charging time	BLSTM, MSCNN, Hybrid DCNN, Hybrid BCNN
Zhu et al. (2022b)	Degradation assessment	The size of the tool wear regarding the cutting number	Traditional machine learning methods depend on empirical design, and their prediction didn't involve a corresponding confidence interval.	The post-probability computation of DBN allows online inference with a confidence interval	Degradation stage	LSTM and its variants, autoencoder, FANN, SVR

- RUL prediction (17 papers)
- Anomaly detection (3 papers)
- Failure prediction (9 paper)
- failure identification (2 papers)
- degradation assessment (2 papers)

Research in this category mainly focuses on RUL/failure prediction, which also implies the capacity of DRNN-based architectures in condition forecasting. The most mentioned challenge that is addressed by DRNN networks is the difficulty in discovering knowledge, correlations, or features about failures or anomalies in long-term temporal dependencies or sequential data. For example (Das et al., 2018), indicated the main challenge in failure prediction is obtaining location information about the anomaly and forecasting accurate lead times by properly processing timestamps among several events across nodes (Mao et al., 2018). consider that the process of bearing degradation is

Table 4
Applications of DRNN networks in predictive maintenance.

Ref.	Task	Output	Filled gap	Advantages or reasons for using the structure	Input signals or data type	Other methods with which this method was compared
Li et al. (2022)	RUL prediction	Bearing health state	The mathematical modeling approach describes real-world complex dynamic systems heavily rely on empirical expert prior knowledge, which is intractable to be obtained.	Due to deep architecture that captures long and short time dependencies, DRNN was considered to perform a better natural ability in long-term prediction tasks.	Vibration signal	Hidden Markov model, CNN
Diez-Olivan et al. (2021)	Anomaly detection	Drift detection	The working circumstance of equipment usually comprise nonstationary dynamics, unexpected alteration, and few extreme positive or negative noisy, which still require a mechanism to be adapted well in the tasks of PHM.	LSTM tend to extract applicable and relevant information from measurement data.	Vibration signal	Stochastic gradient descent-based neural network, Gaussian Naïve Bayes classifier
Netisopakul and Phumee (2022)	Degradation assessment	Temperature and insulation values of motor	The manual maintenance schedule can't satisfy the requirement of specific machine repair, whose deterioration may be caused by severe damage. Also, the setup of a hyperparameter of LSTM may stifle its advance.	DRNN-based approaches may recognize the deterioration point at the early stage, hence, figure out future trends of the life cycle of equipment by digging hidden information from long-term measurement.	Temperature and insulation data	Multi-layer perceptron
Chen et al. (2021)	Degradation assessment	SOH and remaining capacity of the battery	Traditional corrective maintenance and preventive maintenance only fix and replace obvious failure components, which denoted the disadvantage of low reliability and high downtime.	The disadvantages of gradient explosion and vanishing are fixed. Furthermore, DRNN specializes in seizing long-term dependency for nonlinear dynamic systems.	Voltage and current of terminal and charger, temperature	RNN family included RNN, LSTM, GRU and BLSTM
Lu et al. (2021)	RUL prediction	Remaining working cycle of bearing	Existing learning-based methods are still hard to figure out the adaptation toward various operation conditions for mapping nonlinear degradation trends.	LSTM has been widely validated that it can memory worthwhile long-term and short-term system dynamics in back-propagation training.	Vibration data	Quadratic regression model
Mateus et al. (2023)	RUL prediction	RUL of rollers in hot strip mills	The empirical prognosis strategy only finds the relationship between the combination of pressure, contact length, rolling length, etc., with the remaining useful life cycle. However, central factors limited the comprehensive description of actual circumstances.	According to the forget and memory mechanism, LSTM can figure out both sparse and dense time dependencies within a nonlinear dynamic system.	Roller maintenance data	CNN, RNN, DBN
Li et al. (2020b)	RUL prediction	Health conditions of the cutting wheels	In practice, the comprehensive condition of a mechanical component within any equipment may be hard to parse due to partial observation around the surface of the machine. In this context, the inference of degradation becomes complicated.	Deep RNN is widely investigated in that it can effectively mine useful features within sequential data and map those features to the health state of the machine. Also, the attention mechanism is well adapted to the DRNN architecture.	Cutting wheel degradation image	DCNN, NoSupAtt
Li et al. (2022)	RUL prediction	RUL of bearings	The fusion strategy for both time-domain and frequency-domain information enhances the prediction of health state, which is difficult for traditional methods to capture valuable features. Besides, it's much more complicated for a CNN architecture to capture long-term dependency. What's more, most of the existing prognosis strategies didn't figure out the fault occurrence point logically.	Two-dimension LSTM can recognize both long-term and short-term temporal information, as well as deal with spatial feature fusion, such as vertical and horizontal feature fusion	Vibration signals from bearings	DCNN, Hidden Markov models (HMM)
Liu et al. (2021)	RUL prediction	RUL of rolling bearings	Most of the existing research didn't point out that degradation factors may occur at any time in the life cycle. Shallow architecture tends to lose a certain part of useful information due to its nature, and traditional LSTM was found sometimes trapped in an overfitting problem.	Along with the regularization strategy, DRNN architecture figures out the problem of overfitting during the training process of learning long-term dependency	vibration signals from rolling bearings	BPNN, RBF, SVM, DBN, CNN
De Simone et al. (2023)	Failure prediction	Condition of cooling systems	At the early stage, the accumulative deterioration factor affects the rail system steadily then gradually leads	LSTM can forecast the trend of degradation based on gradually changing system dynamics at the	Flow rate of liquid, level of liquid, the	DT, RF, MLP

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Table 4 (*continued*)

Ref.	Task	Output	Filled gap	Advantages or reasons for using the structure	Input signals or data type	Other methods with which this method was compared
			to degradation at a later stage. It's a challenge for deep learning models to capture different time-dependency at different stages.	early stage. Long-term dependencies are memorized over time.	temperature of IGBT, and circuit.	

one of temporal variation. Thus, the research's main challenge was extracting features about faults from temporal information to predict bearings' RUL. Lu et al. (2018) think the main challenge in early fault detection for bearings is capturing information from sequential data correlations. Many other similar challenges, which are caused by the lack of methods for acquiring knowledge from long- or short-term dependencies, are mentioned in the literature, such as dealing with the issue that signals with time lags make the original correlations among sensors unclear (Zhang et al., 2018c), studying long-term time dependencies of the aging features in the degradation evolution of lithium-ion batteries (Zhang et al., 2017), and processing a large number of sequential features (Li et al., 2022).

5.4.2. Result of RQ2: advantages of DRNN networks

The most significant advantage of a DRNN network is its strong ability to discover knowledge or information from long-time sequential data. Through stacking memory cells, the network can, to a certain degree, retain information hidden in the previous time steps in the current time step. Thus, a DRNN network can obtain knowledge from long-time sequences and keep that in memory (Lu et al., 2021; Das et al., 2018) also found that an LSTM network can identify anomalies that may lead to node failures in past phrases and estimate the time of the subsequent failure. For most applications, DRNN is capable of obtaining the best performance among the approaches we proposed in this paper (Maschler et al., 2022). indicated the reason why LSTM became the popular selection of time-series modeling, that it can capture and memorize long-term dependencies which helps with prediction at the early timing. Seo et al. (2018) also stated that DRNN could deal with the problem of long-term dependency prediction by sharing the previous information in the memory cells. Research by (Mao et al., 2018) also validates that DRNN performs well at finding the temporal relevant information in long time-series data since the memory units can fuse various length period information over time. Zhang et al. (2018c) also consider that DRNN has a strong ability to retain and access information, making it possible to deal with long historical trends. In (Zhang et al., 2017), the constructed DRNN successfully detected key features from inputs at an early stage and "kept feature information over a long time, and further discovered the hidden long-term dependencies for RUL prediction."

Other advantages of DRNN mentioned in the studies include their capability "to characterize the system complexity and to translate the raw sensor data to an interpretable health index" (Wu et al., 2018), to build in mechanisms to model time-dependent relations (Lima et al., 2018), and to extract features and capture fault dynamics (Niu et al., 2018).

5.4.3. Result of RQ3: limitations of DRNN networks

As mentioned above, the essential advantage of DRNN is the strong ability to deal with long-time sequential data. We also noticed that most successful applications of DRNN-based architectures in the literature require time-dependent or sequential data as inputs. This situation arises because of the natural structure of DRNN since the use of multiple memory cells in DRNN also requires multiple and continuous time steps in one input sample. Therefore, in predictive maintenance, capturing the phenomenon of potential failure by DRNN requires run-to-failure data (Das et al., 2018; Li et al., 2022; Lima et al., 2017; Lu et al., 2021; Zhang et al., 2018a, 2018b, 2019) or simulation data with equal quality (Lima et al., 2018; Nguyen and Medjaher, 2019; Seo et al., 2018). However, collecting those data in many practical applications is usually difficult, especially for expensive or safety-critical systems. Thus, we consider DRNN only suitable for the cases in which run-to-failure sequential data or failure data collected with continuous timestamps are available. In addition, since every single additional timestamp will raise the number of original dimensions of the input by one, extending the time span of a DRNN will significantly increase the modeling complexity.

5.5. Convolutional neural networks

CNN architecture is commonly considered suitable for learning the data in the form of multi-dimensional arrays or having a grid-like topology. For instance, a color image can be considered as a matrix concatenated by two-dimensional arrays, which include three color channels. Actually, various types of industry data can be viewed as arrays, for example, one-dimensional arrays for signals in sequences, two-dimensional arrays for spectrums or images, and three-dimensional arrays for videos or stereo images. The core idea for CNNs is to leverage the properties of natural signals through specialized architectures with local connections, shared weights, pooling layers, and a series of stages.

Compared with conventional neural networks, CNNs introduced two special layers: convolutional and pooling. In convolutional layers, all units are linked to local patches through feature maps. Units in one feature map share a set of weights, which are also called filter banks. Regarding the mathematic principle, the extracting operation of feature maps is a discrete convolution, which is the origin of the name CNN. Pooling layers can compute the maximum value of a local patch, integrate similar features, and suppress the dimension of the hidden pattern that represents the linear transformation of origin data while keeping invariance to slight shifts and distortions. Table 5 shows the ten primary and typical applications of CNNs in predictive maintenance.

5.5.1. Result of RQ1: main tasks of CNNs

A total of 18 papers successfully applied CNN-based deep learning architectures to achieve specific tasks for predictive maintenance. These tasks include.

- RUL prediction (12 papers)
- failure prediction (3 papers)
- damage, failure, or defect identification (15 papers)
- degradation assessment (1 paper)

We notice that CNN-based deep learning architectures have become a popular research topic and applications that applied in many areas of predictive maintenance. RUL prediction and damage and failure or defect identification are the most researched issues among them. Most research in this category mentioned that the initial target is to extract useful information about failures, damage, or degradation from complex and nonstationary signals (Zhu et al., 2019). For instance (Li et al., 2019c), cast a new light on the challenge of data-driven degradation assessment, they found that the majority problem should be identifying effective and representative details from the raw measurement. Dong et al. (2019) also state that the main challenge in predicting forthcoming faults in bearing degradation processes is to extract useful information

Table 5
Applications of CNNs in predictive maintenance.

Ref.	Task	Output	Filled gap	Advantages or reasons for using the structure	Input signals or data type	Other methods with which this method was compared
Behera and Misra (2023)	RUL prediction	RUL of rolling bearings	The result of data-driven approaches apparently drops to an unsatisfied level as a result of imbalanced data, that contain small scale fault samples.	Pre-training strategy enhances CNN's capability of extracting degradation information over time inside of a limited imbalance training image dataset and overcome the overfitting issue in the training process.	Degradation images converted from vibration signals via Markov transition fields	RNN, Enhanced Encoder-Decoder, Transferable convolutional neural network, DenseNet, ResNet, VGG
Sadoughi and Hu (2018)	Failure identification	Defect types of rolling-element bearings	Existing data-driven approaches rely on the quantity and quality of data, such that there is no physical knowledge has cooperated into the learning strategy.	The specific kernel-based computation strategy of CNN allows physical knowledge-based feature validation between convolution kernels and input data by identifying the local similarities.	Signals including vibration, microphone, and acoustic emission	SVM, ANN
Li et al. (2019b)	RUL prediction	RUL of rolling bearing	Extracting the most relevant representation of system conditions from multi-dimension raw measurement is still the majority challenge of data-driven degradation assessment.	Polymorphic CNN kernels can be designed to extract and generated useful abstract discriminative features, at different levels and scales, then validate by a loss function, from raw sensor measurement after training.	Vibration data	Single low-scale and single high-scale feature extraction
Li et al. (2020a)	Failure identification	Transient pattern of temperature	The transient pattern of practical nuclear reactor and simulations is different apparently because of random noise and uncertainties. The variance between the practical model and the simulation model makes the failure classifier unstable in practical application.	CNN can learn high-level representations automatically, which facilitates further clustering of different patterns.	Temperature, pressure, and flow rate measurements	Self-organizing map, Auto encoder
Yang et al. (2019a)	RUL prediction	RUL of bearing	The effectiveness and generalizability of traditional feature engineering has met the basic requirement of prediction accuracy in most of applications, nevertheless, some relevant detail information may be missed out during traditional feature engineering.	Instead of targeting at a specific domain of raw measurement by using traditional feature extraction approaches, CNN presents the ability of recognize the inherent variance at the normal and failure stage of equipment automatically.	Vibration signals	SVR
Wang et al. (2020)	Failure identification	Failure types of power rig and fan.	Various type of intrinsic features of raw measurement still needs to be explored instead of applying more coarse learning-based methods	The architecture of multi-channel and multi-layer allows CNN to enhance the ability of feature extraction and fusion, in the context of different types of measurement.	Color images converted from multi-sensor vibration signals	DBN, SVM
Cheng et al. (2020)	degradation assessment	degradation of bearings	The complexity of mathematical models limited the prediction performance of model-based approaches. On the other hand, classic machine learning techniques have difficulty in recognizing the degradation pattern and health state, from complicated scenarios.	By using a carefully designed CNN, the receptive field is allowed to capture the degradation pattern from raw measurement at different stages of the working cycle, thus, the result can indicate the health state properly.	Vibration signals	SVR
Diraco et al. (2021)	RUL prediction	RUL of rotating machinery	The information acquired by most of the traditional feature engineering and signal processing techniques needs huge time to be validated, but also represents a limited meaningful pattern that corresponds to degradation assessment under certain circumstances.	CNNs can automatically discover representative features of degradation states from sensor data.	3D scan data	SVR
De Santo et al. (2022)	Failure prediction	Failure types of bearings	Most of existing techniques make precise predictions based on the high-quality well-structured training dataset, whose quality is hard to guarantee and validate in most industrial scenarios.	Adversary CNN architecture tends to introduce a novel mechanism to recognize failure based on generating fake samples from Gaussian distribution.	Encoded images generated from vibration data	LSTM, GRU, XGBoost, and two types of popular CNN.
Wang and Yan (2022)	Failure identification	heating, ventilation, and	Traditional methods usually perform single-fault-level fault diagnosis and single-domain fault	CNN can utilize multiparty data for multi-scale joint modeling to	Sensor data collected from temperature sensors, valve sensors,	LSTM, Light GBM, and GRU

(*continued on next page*)

Table 5 (*continued*)

Ref.	Task	Output	Filled gap	Advantages or reasons for using the structure	Input signals or data type	Other methods with which this method was compared
		air conditioning systems	identification, which cannot simultaneously perform fault diagnosis of different levels.	simultaneously detect and identify multilevel faults	air pressure sensors, etc.	

from complex operation processes and that many unknown conditions may occur alternately. A similar situation is also mentioned in nuclear reactor malfunction identification (Hou et al., 2018), in which the main challenge is to separate various dissimilar patterns into several clusters based on the exploration of distinctive features in the raw measurement.

5.5.2. Result of RQ2: advantages of CNNs

As explained by (LeCun et al., 2015), CNNs are structured to capture distinctive local motifs or information from the data, which are highly correlated or invariant to location. Thus, a CNN can deal with complex imagery scenes and performs well at local and global feature extraction and object detection (Li et al., 2019a). It provides reliable and robust performance in the scenario that involves the multi-scale local and global feature fusion approach, as well as image processing. Research in this category further validates this advantage. For example, according to the experiment by (Yeh et al., 2019), CNN can learn high-level representations automatically, which facilitates further clustering of different patterns in nuclear reactor transients. CNNs can also extract discriminative features from the raw data for prognostics of rolling bearing (Behera and Misra, 2023; Li et al., 2019b; Sadoughi and Hu, 2018).

In (Sadoughi and Hu, 2018), A CNN successfully extracts features that incorporate useful information from physical knowledge about bearings and their fault characteristics from an input signal by quantifying the local similarities between the convolutional kernels and input signal. In (Behera and Misra, 2023), A pre-trained CNN can extract informative degradation features and representations and overcome the limited training data availability problem. Thus, CNN does not require pre-processing and can automatically extract the inherent differences between normal and failure signals, thus reducing the need for prior knowledge, data, and experimentation (Yang et al., 2019a).

5.5.3. Result of RQ3: limitations of CNNs

As mentioned, CNN-based deep learning architectures are structured to capture distinctive local motifs or information from the data, which are highly correlated or invariant to location. The limitation is significant because most successful applications of CNNs require large-scale array data as inputs, such as images, local groups of values, or data with strong relationships between neighboring variables.

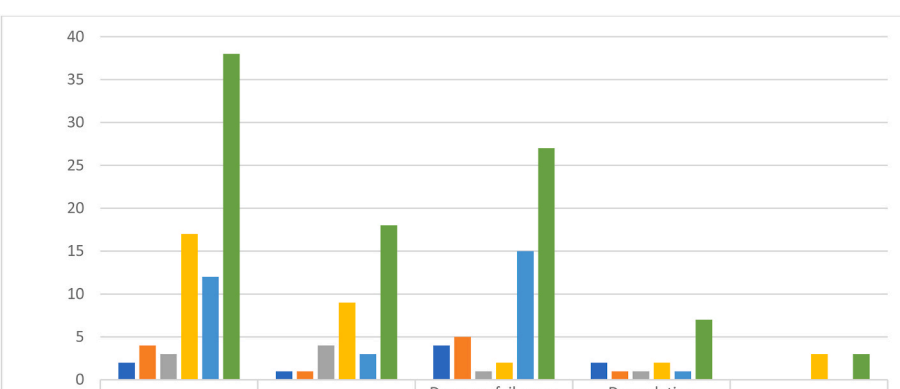
Among the 32 studies that used CNNs, 10 directly applied images as the inputs, and 13 of them first transformed sensor signals into a matrix and then used this as the input. Therefore, we consider that the type of input data restricts CNN-based deep learning architectures.

6. Discussion

In the previous section, our results indicate that the top three tasks of deep learning methods for predictive maintenance are RUL prediction; damage, failure, or defect identification; and failure prediction (as seen in Fig. 3). LSTM networks and CNNs are the most widely applied deep learning architectures in RUL prediction and damage, failure, or defect identification, respectively, because of the ability to discover knowledge about potential failures or anomalies from long-time sequential data and to capture specific local information from data. However, we also notice that due to the diversity in practical applications, such as the complexity of the issues and properties of the input data, when to use which learning architecture needs guidelines. To enhance the explanation of the guidelines in section 6.2, a detailed description of the survey result is provided regarding the deep learning technique specification. In Section 6.2, based on our literature review results, we will provide guidelines for researchers and practitioners to select proper deep learning architecture in predictive maintenance.

6.1. Guidelines for selecting deep learning architectures

According to the above theoretical interpretations and reviewed applications, we divided the decisive factors in choosing deep learning



	RUL prediction	Failure prediction	Damage, failure or defect identification	Degradation assessment	Anomaly detection
DNN	2	1	4	2	0
SAE	4	1	5	1	0
DBN	3	4	1	1	0
DRNN	17	9	2	2	3
CNN	12	3	15	1	0
Sum	38	18	27	7	3

Fig. 3. Summary of architecture application areas.

architectures into four dimensions: (1) target of the task, such as degradation assessment, RUL prediction, health estimation, and damage identification; (2) the features that may need to be extracted, including time dependency features, complex representations, discriminative features, and hierarchical representations; (3) the input data, including coupled or disordered levels, data types, and relationships between neighboring variables; and (4) special demands on analysis capability, including the capabilities to deal with temporal information, complex environments, and complex imagery scenes. The dimensions and decisive factors are shown in Fig. 4.

The guidelines for selecting suitable deep learning architectures for predictive maintenance based on the four dimensions are summarized as follows.

(1) DNNs with fully connected hidden layers and backpropagation usually perform outstanding RUL prediction, failure prediction, and degradation mapping when the input dimension is not high, historical data cover the target condition, and the modeling or environment is complex. The computation complexity of the optimization of DNN is relatively lower than the other architectures, meanwhile, the final results of the convergence were stable in the context of setting up appropriate hyperparameters. Overall, for most applications, DNNs can at least provide a fundamental robust benchmark for the further development of deep learning architectures.
(2) SAEs usually have a strong ability in fault characteristics mining to automatically extract hierarchical representation features and information for failure identification through sensor data and divide them into various levels afterwards, as well as to identify discriminative information under a large-dimensionality environment. The implementation of SAE sometimes requires a strong understanding of domain knowledge to justify the effective hidden variables, which were commonly used to denote the health precursor of the system. Due to the similar structure, the computation complexity of SAEs is close to the DNNs.
(3) DBNs can discover significant information from coupled or disordered data with good generalization ability because of the energy-based structure. Thus, DBNs can deal with failure prediction or health assessment in an environment where the collected data are highly coupled, disordered, or even beyond the range of historical data. In most cases, DBN was a cost-effective solution for the dataset that presented localized and hierarchical high-dimension features. Compared to DNNs and SAEs, the cost of computation could be much higher because of the pre-training process.
(4) DRNN were developed to discover temporal information in sequential data. The stacked memory cells inside the network can help keep long-term historical information during operations, which allows LSTMs to mimic time series and predict specific parameters for predictive maintenance. In practice, the implementation of DRNNs was much more difficult than the others because it was hard to figure out the trade-off between the capacity of memory and the overfitting issue. The cost of computation is relatively higher than the previous methods. As reported by (Zhao et al., 2019), "most machinery data is obtained from sensor data, which are based on time series in nature." We believe that this is why LSTM networks are the most widely applied method for failure or RUL prediction in our review.
(5) CNNs generally have a solid capacity to discover knowledge from array data, including groups of values and images, for different types of targets in predictive maintenance. Owing to the two-dimension architecture, the computation complexity of CNN is the highest among these architectures, however, depth-wise separable CNN could be a better option for developers. The setup of the hyperparameter of CNN, such as kernel size and dilation, should be considered carefully based on the analysis of the data, for example, defining the size of local features. In addition, due to their ability to discover complex representations through the convolutional layers, CNNs could identify both local and global features to estimate and predict the state of the equipment.

6.2. Dataset requirements

The aforementioned introduction of the requirements of data acquisition has indicated several significant aspects related to the experiment setup and sensor techniques. Some popular open-source datasets such as C-MAPSS and IEEE 2012 Prognostics and Health Management Data Challenge dataset, were capable of providing sufficient measurement from various perspectives such as data dimensions and sampling rates. These run-to-fail datasets covered the observation of the life cycle of the equipment so that a comprehensive health state estimation could be inferred by deep learning models. In the research given by (Li et al., 2019e), the measurement of the bearing vibration signal was set up in 3 directions separately, with the sampling rate at 4096Hz to guarantee the observation of frequency multiplier information. To

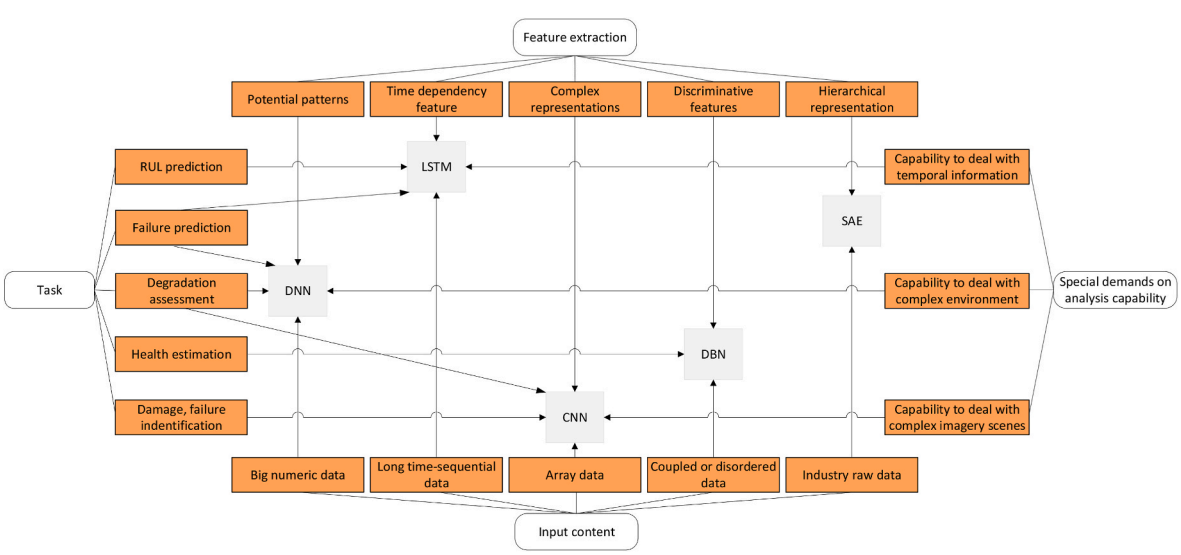


Fig. 4. Roadmap for selecting proper deep learning architectures.

obtain a comprehensive classification of the failure of the bearing, 3 types of failure were injected into the bearing during the operation. Another effective run-to-fail dataset was established based on the data acquisition system that involved the National Instrument board and LabVIEW software on the test rig (Deutsch et al., 2017b). The authors introduced the utilization of filters and the configuration of the software interface (Haris et al., 2021). introduced the utilization of the datasets published by (Zhou et al., 2020). The control strategy of the supercapacitor aging experiment includes setting up the reference voltage with respect to different temperature conditions, which is regarded as the main factor that led to the deterioration of the supercapacitor. Similarly, the run-to-fail dataset for the health assessment of the battery was also conducted by applying different settings of temperature (Chen et al., 2021).

6.3. Comparison with related work

As mentioned in Section 3 and Section 4, recently, especially in the past four years, deep learning–driven approaches are increasingly applied to various types of tasks for predictive maintenance due to their comprehensive and successful applications in the AI domain. During the literature review, we also noticed some related work that aims to review the development of computer-aided maintenance. For example (Peng et al., 2010), reviewed various technologies and algorithms that have been applied to machine prognostics and classified them into physical models, knowledge-based models, data-driven models, and combination models. Some shallow learning methods, such as multilayer perceptron neural networks and self-organizing feature maps, are also studied and introduced in that paper (Prajapati et al., 2012). provided an overview of recent developments and applications in condition-based maintenance, focusing on maintenance strategies. The work by (Khan and Yairi, 2018) also reviewed deep learning applications in system health management. This study focuses on data classification and feature extraction problems without evaluating the effectiveness and guidelines for selecting or designing suitable architectures. Our research covered new studies from 2017 to 2023. To the best of our knowledge, our work is the first systematic survey that summarizes and details the application scopes, advantages, and limitations of popular deep learning approaches and guidelines for selecting suitable architecture for predictive maintenance.

6.4. Threats to validity

In this section, we discuss potential threats to the validity of our survey.

6.4.1. Search strategy

The main threat in this step is missing or excluding relevant literature. To mitigate this threat, we applied two strategies. In the first place, our search covered synonyms for possible keywords that may have been included in various studies. The applied keywords and synonyms were designed to narrow down the scope and collect relevant literature. Secondly, we conducted an extensive snowballing process for the references of the selected primary studies to ensure coverage. The first and the third authors discussed and cross-validated the searched keywords.

6.4.2. Study selection

Selecting papers and excluding others could also threaten validity, as the selection process may be affected by the subjective judgment of the authors. To mitigate this threat, we predefined the selection criteria and strictly followed the review protocol, with all authors contributing to the validation. The first author reviewed the title, keywords, and abstract of all the searched 248 articles and extracted the main tasks, applied methods, and objects for selection. The second and third authors validated and improved the selection result by considering the review scope and predefined research questions. For example, we included studies that involve deep learning–based methods to predict potential failures for maintenance service. Still, we excluded articles focusing on route decision modeling or measurement compensation.

6.4.3. Data extraction

In this stage, the main threat is the possibility of missing information from the primary studies. To minimize this threat, the first author was responsible for designing the data extraction strategy and conducting the strategy to extract information from the selected literature to answer the research questions. The third author improved and confirmed the coverage of the data categories, while the second author verified the extracted data. All authors continuously discussed the data extraction issues to avoid the first author's bias in data extraction.

6.4.4. Data synthesis

To avoid the risk of researcher bias during the data analysis process, thematic synthesis steps are followed. In this study, the extracted data were synthesized by the first author. Then, the second and third authors discussed the data to validate it. The final data synthesis was agreed on by all authors.

7. Conclusion and future work

In recent years, with the increased adoption and development of AI and IoT technologies, deep learning architectures have been widely applied in predictive maintenance and have shown outstanding performance in many applications. To some extent, deep learning exhibits substantial promise for solving complex challenges in engineering applications across a multitude of industries. On the other hand, we also indicated that it's still a challenge that engineers need to consider the usage of an appropriate DL architecture carefully based on the data type in a practical application. Besides, according to the reviewed paper, it is worth noticing that the interpretability and reliability of NN could be enhanced by introducing self-adaptive and physics-assisted mechanisms. This paper presents a survey of 106 primary studies focused on applying deep learning architectures in predictive maintenance. Based on the literature review, this study proposed a classification of five deep learning architectures that are widely applied to various tasks in predictive maintenance. These tasks have emerged as one of the most significant parts of predictive maintenance of the industrial equipments and their component. Eventually, it is of considerable importance to enhance the reliability of industrial equipment. At the end of each subsection of section 5, we also summarize the advantages and limitations of each deep learning approach and give guidelines for researchers and practitioners for selecting suitable deep learning architectures for predictive maintenance. In section 6, we indicate the most relevant advantages for each approach based on the technical analytic of their nature, within their area of expertise. In other words, the structure of each approach tends to leverage different underlying features that exist in the data. Time-frequency domain information, hierarchical information and complex representation stand out as the primary factors that engineers need to consider carefully. We need to point out that a certain proportion of reviewed papers have indicated that DL architectures will be an effective evolution of traditional signal processing approaches. Above all, as the predictive maintenance strategy has been clarified by organizing specific DL architecture to corresponding task, we have addressed the challenge of identifying the decisive factor of suggestion for the engineers who want to develop DL applications.

As we listed in Section 5, deep learning–driven approaches are widely and successfully applied to various tasks and targets for predictive maintenance. Most primary studies in the literature detailed their application scenarios, explained the selection of deep learning architecture, and validated the advantages of deep learning theoretically and practically based on effective benchmark comparison experiments. However, very few studies address the development of transfer learning in predictive maintenance. During the literature review, we noticed that

only two articles introduced transfer learning in their research to extract features about the flank wear of cutting tools (Sun et al., 2019) and to identify degradation from other data sets in the same motor (Wen et al., 2019). The abovementioned studies applied transfer learning to the same content structures and tasks. As mentioned by (Ding et al., 2016), transfer learning aims to unite knowledge from different fields, which enables leveraging validated knowledge from other domains or tasks when the available data for the target are limited or not strong enough. From the authors' perspective, a quantitative or qualitative lack of data is a common dilemma in practical predictive maintenance applications. Thus, validating the feasibility and effectiveness of transfer learning in predictive maintenance applications could be a future research direction in this field.

Our future work may also focus on studying the advantages of each deep learning architecture using theoretical analysis to further validate the improved performance or to solve bottlenecks in predictive maintenance based on deep learning methods. To encourage the development of the deep learning architectures in the practical industrial scenario, more comprehensive review towards the deep learning enabled predictive maintenance for specific applications, such as production lines, and power system will be the vision of our future work. One of the drawbacks of current development of deep learning is lacking interpretability, in this case, the deep learning architecture that present consistent with physic rules will be definitely meaningful to the advancement of practical applications. On the other hand, regarding the advancement of deep learning, the applications that involved more newly developed integrated architectures and algorithms could be explored in terms of the reliability framework of industrial systems. To enhance the performance of online learning in the industrial applications, the efficiency of the computation could be significant to the common operation, thus, we intend to explore more effective solutions to reduce the computation complexity of deep learning architectures which should maintain the prediction accuracy.

CRediT authorship contribution statement

Zhe Li: Writing – original draft, Methodology. **Qian He:** Writing – review & editing, Formal analysis, Data curation. **Jingyue Li:** Validation, Investigation, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Zhe Li reports financial support was provided by Science and Technology Commission of Shanghai Municipality. Zhe Li reports a relationship with Shanghai Electric Group Co Ltd that includes: employment.

Data availability

No data was used for the research described in the article.

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